

Samuel Titus Oluwatimileyin

Backend Developer · Node.js · PostgreSQL · REST APIs · DevOps

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SUMMARY

Self-directed backend developer with a BSc in Computer Science and a track record of shipping documented, production-grade APIs. I build systems that solve real problems — from a fully deployed fintech platform with live observability and event streaming, to secure auth pipelines, async job queues, and NIBSS-integrated banking flows. Proficient in Node.js, Express, PostgreSQL, Redis, Docker, and GitHub Actions CI/CD. Open to junior backend and software engineering roles.

TECHNICAL SKILLS

Languages: JavaScript (primary), TypeScript, Python

Runtime: Node.js

Frameworks: Express.js

Databases: PostgreSQL, MongoDB, Redis

Queue / Jobs: BullMQ, Kafka (KRaft mode, kafkajs), exponential backoff, background worker architecture

Auth & Security: JWT (access + refresh tokens), bcrypt, RBAC, Zod input validation, Helmet, CORS, rate limiting

Observability: Prometheus (prom-client), Grafana, Morgan

DevOps / Infra: Docker, Docker Compose, multi-stage Dockerfiles, GitHub Actions CI/CD, Railway cloud deployment

Integration: NIBSS / NibssByPhoenix (BVN/NIN verification, inter-bank transfers), Telegram Bot API, axios

Tooling: Git, Postman, node-cron, connection pooling (pg), Jest, Supertest

Concepts: REST API design, async job processing, multi-role authorization, state machine flows, middleware architecture, domain-modular project structure, ACID transactions, cache invalidation, event-driven architecture

BACKEND PROJECTS

Nigerian Banking API (Production) Node.js · TypeScript · Express · PostgreSQL · Redis · Kafka · Docker · Prometheus · Grafana · Railway

github.com/saintamadeus/nigerian-banking-api Live: nigerian-banking-api-production.up.railway.app

- **Production-deployed REST API** on Railway with managed PostgreSQL and Redis — JWT-authenticated endpoints for account management, fund transfers, and full transaction history; live and publicly accessible
- **ACID-safe transactions:** PostgreSQL FOR UPDATE row locking prevents race conditions on concurrent fund transfers; all writes wrapped in explicit BEGIN/COMMIT/ROLLBACK with connection pooling via pg
- **Security hardened:** IDOR prevention via user_id-scoped SQL on every query, bcrypt (salt rounds 12), Helmet security headers, and tiered rate limiting (100 req/15 min general, 10 req/15 min on auth routes)
- **Redis caching with graceful degradation:** account reads cached with TTL and invalidated on every transaction; Redis failure is non-fatal — private helpers wrap all cache ops in try/catch so a Redis outage never corrupts a committed database write
- **Kafka event streaming (KRaft mode):** publishes TRANSACTION_COMPLETED events to transaction.completed topic after every successful DB COMMIT; standalone consumer.ts subscribes and processes events; Kafka failure is non-fatal — app continues without it if KAFKA_BROKER is unset
- **Full observability stack:** Prometheus scrapes /metrics every 15 seconds via Railway internal network; Grafana dashboard with HTTP request rate, total requests by route/status, and memory usage panels
- **Production-grade Docker setup:** multi-stage Dockerfile (builder → production), docker-compose with health checks on all four services (PostgreSQL, Redis, Kafka, app), named volumes, and automated schema creation via init.sql
- **CI/CD pipeline:** GitHub Actions runs 11 Jest + Supertest integration tests on every push against a dedicated test database; npm ci for reproducible builds; pinned action versions for stability

Digital Banking API (TIT Bank) Node.js · Express · PostgreSQL · JWT · NIBSS Integration

github.com/saintamadeus/digital-banking-api-nibss

- REST API for a full digital banking system — customer registration, JWT auth, identity verification, account management, and intra/inter-bank fund transfers
- NIBSS integration: integrates with NibssByPhoenix core banking infrastructure for BVN and NIN identity verification; NIBSS tokens are fetched and cached server-side automatically so the calling client never handles credentials
- Transfer flow: name enquiry endpoint verifies recipient before any funds move; intra-bank and inter-bank transfers both supported; transactions stored with unique reference IDs for status lookup and audit
- Data isolation: customer_id enforced at the SQL query level on every data retrieval — customers cannot access each other's transaction history regardless of auth state
- Domain-modular structure: auth, onboarding, accounts, transactions — each self-contained with its own routes, controller, and DB logic

Food Supply Chain Tracking API Node.js · Express · PostgreSQL · JWT · Helmet

github.com/saintamadeus/food-supply-chain-api

- Multi-stakeholder REST API tracking food products from producers to consumers across six roles: admin, producer, supplier, warehouse manager, distributor, and retailer — each with distinct authorized actions
- Access + refresh token pattern: separate short-lived access tokens and longer-lived refresh tokens; logout invalidates the refresh token server-side
- Full audit trail: every batch has a queryable event trail (/batches/:id/trail) recording every status transition across its lifecycle
- State machine flows: batch (created → ready → in transit → stored → sold/expired), shipment (pending → in transit → delivered/cancelled), and order (pending → approved → fulfilled/unavailable) — all enforced at the API layer
- Postman documentation published; Helmet + CORS configured; modular folder structure by domain

Job Queue API Node.js · Express · BullMQ · Redis · PostgreSQL

github.com/saintamadeus/job-queue-api

- Async job processing system that fully decouples task submission from execution using BullMQ backed by Redis
- Retry logic: exponential backoff — failed jobs retry at 5s, 25s, and 125s intervals to handle transient failures without hammering dependent services
- Full job lifecycle tracked in PostgreSQL (pending → processing → completed/failed); clients poll status via REST; payloads stored as JSONB for schema-free extensibility

Auth API Node.js · Express · PostgreSQL · JWT · bcrypt · Zod

github.com/saintamadeus/auth-api

- Production-ready authentication API — registration, login, protected routes — with bcrypt password hashing, JWT auth middleware, Zod request validation, and PostgreSQL connection pooling

Daily Briefing API Node.js · Express · PostgreSQL · node-cron · Telegram Bot API

github.com/saintamadeus/daily-briefing-api

- Personal automation API — sends a consolidated Telegram message at 7:00 AM Lagos time every morning via cron; CRUD API manages reminders with partial update logic using SQL COALESCE

EDUCATION

Bachelor of Science — Computer Science

Federal University Lokoja, Kogi State · 2024

Capstone projects: Food Supply Chain Tracking System (Group 17), SMS Spam Filtering System (Python/ML), Email Phishing Detection System (Python/ML), Blockchain Drug Tracking System